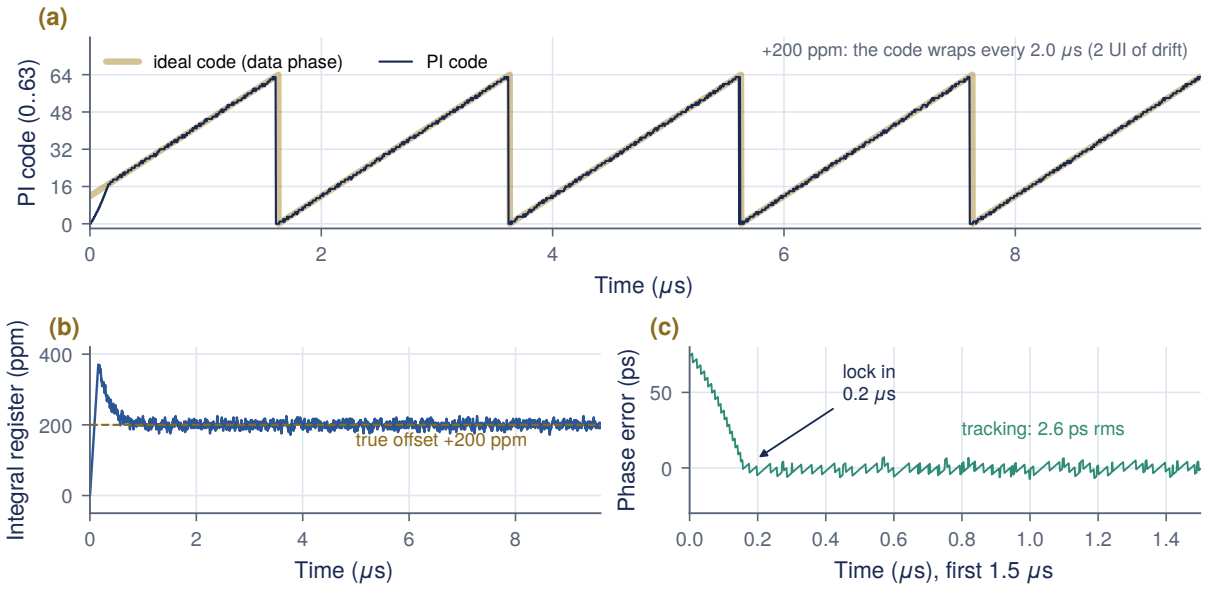
(a)
ideal code (data phase)
PI code
+200 ppm: the code wraps every 2.0 μs (2 UI of drift)
PI code (0..63)
64
48
32
16
0
0
2
4
6
8
Time (μs)
(b)
Integral register (ppm)
400
200
0
true offset +200 ppm
0
2
4
6
8
Time (μs)
(c)
Phase error (ps)
50
0
lock in
0.2 μs
tracking: 2.6 ps rms
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time (μs), first 1.5 μs